Linear Optimal Topic Transport for Scalable Document Similarity

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Abstract

Document classification plays a critical role in organizing and analyzing large-scale textual data. Despite advancements in document similarity methods, challenges persist in balancing computational efficiency with classification accuracy. In this paper, we introduce the Linear Optimal Topic Transport (LOTT) method, which leverages linearized optimal transport embedding for efficient and robust document classification. LOTT builds on topic modeling and embedding techniques, achieving substantial computational gains while maintaining competitive accuracy across diverse datasets. Our experiments demonstrate that LOTT outperforms existing OT-based methods in scalability and efficiency. It was then compared with different variants of BERT models on the same datasets. We find that LOTT achieves comparable performance with an average test error only 0.051 higher than SBERT, while demonstrating substantial computational efficiency gains of 33× speedup, making it a reliable choice for large-scale classification tasks.

1 Introduction

Document classification is a fundamental task in natural language processing (NLP) that involves categorizing text documents into predefined classes. This process is critical in applications such as organizing news articles, filtering spam, and sentiment analysis. As the volume of digital text grows, the need for efficient and accurate document similarity measurement becomes increasingly pressing. Traditional methods struggle to maintain efficiency and scalability with large datasets, and this growing data demand calls for methods that balance accuracy with computational efficiency.

Contributions. We introduce linear optimal topic transport (LOTT) for measuring document similarity and evaluate it through classification tasks. Our method achieves **high computation speed**, outperforming other optimal transport methods in efficiency while maintaining scalability, since LOTT enables distance calculations in Euclidean space; has **comparable accuracy** to hierarchical optimal transport, BERT embeddings, and other baselines in k-NN classification; and features a **parametrized design** that balances computation speed and test accuracy, providing flexibility to tune and adapt to low-data scenarios.

2 Related Work

Early document representations relied on frequency-based statistics: bag-of-words (BOW) models used raw term counts, TF-IDF (Luhn, 1957) adjusted for term rarity, and latent semantic indexing (LSI) (Deerwester et al., 1990) projected into a lower-dimensional concept space. Latent Dirichlet Allocation (LDA) (Blei et al., 2003) modeled documents as mixtures of latent topics, providing interpretability and dimensionality reduction, with similarity typically measured via cosine distance.

Transformer-based models such as BERT (Devlin et al., 2019) and its variants have since achieved strong performance through contextualized embeddings, though at substantial computational cost. Optimal transport has provided a parallel mathematical framework: Word Mover's Distance (WMD) (Kusner et al., 2015) framed document similarity as a transport problem over word embeddings, and supervised extensions (Huang et al., 2016) adapted this for labeled tasks. Topic Mover's Distance (Wu and Li, 2017) and random-feature approximations (Wu et al., 2018) reduced complexity but sacrificed interpretability. Yurochkin et al. (2019) proposed hierarchical optimal transport (HOTT), which uses topics as distributions over words and computes pairwise WMD between topics as the ground cost. While HOTT reduces the transport support size, it still requires solving $\binom{|D|}{2}$ pairwise problems, limiting its scalability to large corpora.

3 Background

3.1 Optimal Transport

Optimal transport (OT) theory, first introduced by Gaspard Monge in 1781 and rigorously formulated by Leonid Kantorovich in 1942, offers a systematic framework for comparing probability distributions by quantifying the cost of transforming one distribution into another. In this work, we focus on the discrete setting of OT and refer the reader to (Solomon, 2018) for detailed theory and application. Consider two finite sets of points, $\mathcal{X} = \{x_1, \ldots, x_n\}$ and $\mathcal{Y} = \{y_1, \ldots, y_m\}$, equipped with discrete probability distributions $\mu \in \triangle^n$ and $\nu \in \triangle^m$, where $\triangle^n$ and $\triangle^m$ are the n- and m-dimensional probability simplices, respectively. We assume that each element $x_i \in \mathcal{X}$ and $y_j \in \mathcal{Y}$ has an associated probability mass μ_i and ν_j, and that $\sum_{i=1}^{n} \mu_i = 1$ and $\sum_{j=1}^{m} \nu_j = 1$.

In the classical OT formulation, we define a cost matrix $C = (c_{ij}) \in \mathbb{R}^{n \times m}$, where c_{ij} represents the cost of transporting mass from point x_i to point y_j. The OT problem then seeks a coupling matrix $\gamma = (\gamma_{ij})$ that redistributes mass from $\mathcal{X}$ to $\mathcal{Y}$, minimizing the total cost. Then we can define the 1-Wasserstein distance:

$$W_1(\mu, \nu) = \min_{\gamma \in \Gamma_{p,q}} \sum_{i,j} c_{ij} \gamma_{ij}$$

subject to the constraint

$$\Gamma_{p,q} = \{\gamma \in (\mathbb{R}_+)^{n \times m} : \gamma \mathbb{1}_m = \mu, \quad \gamma^T \mathbb{1}_n = \nu\}.$$

In this formulation, γ_{ij} denotes the amount of mass moved from x_i to y_j. Once solved, this problem provides both an optimal matching between the supports of the two distributions and a minimal transport cost.

3.2 Optimal Transport for Document Representation

Word Mover's Distance. Kusner et al. (2015) represent documents as normalized bag-of-words (nBOW) vectors and measure similarity via discrete OT, using GloVe embedding distances as the ground cost. Formally, $WMD(d^{k_1}, d^{k_2}) = W_1(d^{k_1}, d^{k_2})$ where the cost $c_{ij} = \|e_i - e_j\|_2$ is the Euclidean distance between word embeddings $e_i, e_j \in \mathbb{R}^m$.

Hierarchical Optimal Transport. Yurochkin et al. (2019) encode each document as a distribution over LDA-derived topics, with pairwise topic distances computed via WMD. The HOTT distance is:

$$HOTT(d^{k_1}, d^{k_2}) = W_1\left(\sum_{s=1}^{|T|} \bar{d}_s^{k_1} \delta_{t_s}, \sum_{s=1}^{|T|} \bar{d}_s^{k_2} \delta_{t_s}\right)$$

where δ_{t_s} is the Dirac delta at topic t_s and $\bar{d}_s^k$ is document k's proportion of topic s.

4 Problem Formulation

WMD effectively captures semantic similarity, but its computational complexity, scaling with $O(|D|^2 n^3 \log n)$, limits its applicability in large datasets. Even with approximations like Word Centroid Distance (WCD) and Relaxed Word Mover's Distance (RWMD), only partial reductions in computational cost are achieved, making WMD challenging to use for large-scale document classification.

HOTT addresses some of these limitations by combining topic modeling with word embeddings, reducing computational costs by structuring the transport problem over a smaller topic space. However, it fails to scale to large corpus: given $|D|$ documents, calculating pairwise Wasserstein distances requires solving $\binom{|D|}{2}$. This is computationally inefficient when applying to algorithms like k-NN, where distances must be computed between each training and testing data. To address these scalability challenges, we propose **Linear Optimal Topic Transport (LOTT)**, a method designed to significantly reduce the computational load in OT-based document classification tasks.

Linear Optimal Transport (LOT) (Wang et al., 2012) was introduced to reduce the high computational cost of calculating pairwise Wasserstein distances. Assuming the existence of an optimal transport map, LOT embeds each target distribution into L^2 -space by computing the optimal transport map from a fixed reference distribution to the target distribution. This linearization projects onto the tangent manifold at the reference, approximating Wasserstein distance via Euclidean distance.

With LOT, we only need to compute $|D|$ optimal transport maps instead of $\binom{|D|}{2}$, and the pairwise distances are calculated in Euclidean space, significantly speeding up computations.

Definition 4.1 (Linear Optimal Transport Embedding)**.** Let σ be a fixed measure in $P_2(\mathbb{R}^n)$. We define the Linear Optimal Transport (LOT) embedding, F_σ, which maps a measure in $P_2(\mathbb{R}^n)$ to a function in $L^2(\mathbb{R}^n, \sigma)$:

$$F_\sigma(\mu) = T_\sigma^\mu, \quad \mu \in P_2(\mathbb{R}^n).$$

This mapping takes a probability measure μ and assigns it the optimal transport map T_σ^μ, which pushes the reference measure σ to μ.

From (Moosmüller and Cloninger, 2021), we have the following theorem.

Theorem 4.1. *Let $\sigma, \mu \in \mathcal{P}_2(\mathbb{R}^n), \sigma, \mu < \lambda$. Let $R > 0, \epsilon > 0$. For $g_1, g_2 \in \mathcal{G}_{\mu,R,\epsilon}$ and σ the Lebesgue measure on a convex, compact subset of $\mathbb{R}^n$, we have*

$$0 \leq \|F_\sigma(\mu) - F_\sigma(\nu)\|_\sigma - W_2(\mu,\nu) \leq C\epsilon^{\frac{2}{15}} + 2\epsilon$$

where g_1, g_2 pushforward reference to μ, ν and $\mathcal{G}$ is the family of all pushforward functions.

With LOT embedding, we may approximate the Wasserstein distance using the linear optimal transport distance in L^2-space. To measure document similarity, we define the LOTT distance for document d^{k_1} and d^{k_2} as the following

$$LOTT(d^{k_1}, d^{k_2}) = \|F_\sigma(\bar{d}^{k_1}) \ - F_\sigma(\bar{d}^{k_2})\|_\sigma$$

where $\bar{d}^k$ represents the topic distribution for document d^k in $\mathbb{R}^m$, the word vector embedding space. This topic distribution is generated by LDA, with each topic represented as the barycentric projection of word embeddings.

5 LOTT pipeline

5.1 Preprocessing and Topic Modeling

Before computing document similarities, we preprocess the text data and derive a distribution over latent topics for each document. The main steps include:

Vocabulary pruning and stemming. We begin by refining the vocabulary. Stop words, very short words, and words that appear too infrequently are removed. We then apply stemming to group together words that share a common root. This process yields a cleaner, more compact vocabulary. Additionally, we ensure every remaining word has a corresponding embedding, often by replacing its vector representation with a suitable aggregate (e.g., the mean) of the embeddings of its related terms.

Document embeddings. With a pruned and stemmed vocabulary, each document is represented by a bag-of-words vector. We can further transform these into embedded representations by assigning each word its associated embedding vector. These embedded document representations serve as input for topic modeling.

Topic modeling. We apply a Latent Dirichlet Allocation (LDA) (Blei et al., 2003) to uncover a set of latent topics that explain the structure of the documents. The LDA model learns topics in a corpus: $T = \{t_1, \ldots, t_{|T|}\}$, where each t_s is a distribution over the vocabulary. We can then obtain the topic embeddings by taking the barycentric projection of word embeddings. Formally, let $E \in \mathbb{R}^{d \times V}$ be the word embedding from $GloVe$ where d is the dimension of word vectors and V is the size of vocabulary, and let $\bar{t}_s \in \mathbb{R}^V$ be the empirical topic distribution over words, then each topic embedding is computed as $e_{t_s} = E\bar{t}_s$.

In addition, LDA provides topic proportions for each document: $\bar{d}^k = \{\bar{d}_1^k, \ldots, \bar{d}_{|T|}^k\}$, where $\bar{d}_s^k$ is the proportion of topic s in k-th document. Thus, each document is now represented as a mixture of a small number of topics.

After these steps, each document is characterized by a distribution over a small number of interpretable latent topics in the semantic space, setting the stage for efficient document similarity computations.

5.2 Constructing LOT Embeddings for Documents

In this section, we detail the procedure for transforming a document's topic distribution into a linear optimal transport (LOT) embedding. Let each document be represented by a discrete measure $\bar{d}^k$ supported on $|T|$ topic embeddings in $\mathbb{R}^d$. The probability mass of $\bar{d}^k$ over each topic reflects that document's composition in terms of latent topics.

Each document $\bar{d}^k$ is a discrete measure over $|T|$ topic embeddings in $\mathbb{R}^d$, where the probability mass reflects topic composition, and the topic space is endowed with a Wasserstein metric so that distances between topics reflect their transport cost. The LOT embedding is then constructed through the following steps:

1. **Reference Distribution:** Select a fixed reference distribution σ over the same set of $|T|$ topic embeddings. The reference σ serves as a common baseline against which we measure all other documents. Conceptually, σ is a known "anchor point" in the topic space.

2. **Optimal Transport Plan:** To relate the document $\bar{d}^k$ to the reference σ, we solve an optimal transport problem. Given a precomputed cost matrix of pairwise Wasserstein distances between topics, we obtain:

$$T_\sigma^{\bar{d}^k} \in \mathbb{R}^{|T|\times|T|},$$

which minimizes the total transport cost of moving mass from σ to $\bar{d}^k$. The matrix $T_\sigma^{\bar{d}^k}$ encodes how each unit of probability mass in σ should be distributed across the topics in $\bar{d}^k$.

3. **Barycentric Projection for Embedding:** With the transport plan $T_\sigma^{\bar{d}^k}$, we compute a barycentric projection to obtain the LOT embedding of the document. For each topic in the reference σ, we use the corresponding row in $T_\sigma^{\bar{d}^k}$ as weights to form a weighted average of the target document's topic embeddings in the original word embedding space:

$$z_i = \sum_{j=1}^{|T|} (T_\sigma^{\bar{d}^k})_{ij} \cdot e_{t_j},$$

where $z_i \in \mathbb{R}^d$ is the barycenter for the i-th reference topic.

4. **Final LOT Embedding:** By performing this barycentric projection for all T topics in the reference σ, we obtain T vectors in $\mathbb{R}^d$. Stacking these vectors together yields a LOT embedding for the document in $\mathbb{R}^{|T|d}$:

$$\mathrm{LOT}(\bar{d}^k) = [z_1, z_2, \ldots, z_{|T|}] \in \mathbb{R}^{|T|d}.$$

This representation linearizes the Wasserstein space of topic distributions. The hierachical optimal topic transport distance could be approximated in the euclidean space, enabling more efficient computations and comparisons in downstream tasks.

5.3 Relation to WMD

When the number of topics is exactly the size of vocabulary, Yurochkin et al. (2019) argue HOTT is equivalent to WMD. When the number of topics is much smaller than the vocabulary size, assuming each document is uniquely represented as a linear combination of topics, WMD is upper-bounded by HOTT plus the loss introduced by topic modeling.

$$\begin{aligned} WMD(d^{k_1}, d^{k_2}) \le HOTT(d^{k_1}, d^{k_2}) + \mathrm{diam}(X)\cdot \\ \left[\sqrt{\frac{1}{2}KL\Big(d^{k_2} \,\Big\|\, \textstyle\sum_{s=1}^{|T|} \bar{d}_s^{k_2} t_s\Big)}\right. \\ \left. + \sqrt{\frac{1}{2}KL\Big(d^{k_1} \,\Big\|\, \textstyle\sum_{s=1}^{|T|} \bar{d}_s^{k_1} t_s\Big)}\right] \end{aligned}$$

The LOTT–HOTT gap arises because LOT linearizes the curved Wasserstein geometry into a flat Euclidean space, introducing approximation error along geodesics.

We have following theorem from (Mérigot et al., 2020).

Theorem 5.1. *Let ρ be the Lebesgue measure on a compact convex subset $\mathcal{X}$ of $\mathbb{R}^d$ with unit volume, and let $\mathcal{Y}$ be a compact convex set. Then, for all $\mathcal{P}(\mathcal{Y})$ and all $p \ge 1$, $W_2(\mu,\nu) \le \|T_\mu - T_\nu\|_{L^2(\rho)} \le C W_p(\mu,\nu)^{\frac{2}{15}}$,*

where the constant C depends on d, $\mathcal{X}$ and $\mathcal{Y}$.

In our case, HOTT essentially is a Wasserstein distance, while LOTT approximates this distance using LOT embedding. Building on this and recent established bounds for LOT (Mérigot et al., 2020; Moosmüller and Cloninger, 2021; Delalande and Merigot, 2023), if we assume that each document distribution is a pushforward of a fixed reference distribution through shifting and scaling—meaning there exists an underlying reference document that generates all documents via these transformation—then we can derive the following bounds for LOTT:

$$\begin{aligned} HOTT(d^{k_1}, d^{k_2}) &\le LOTT_\sigma(d^{k_1}, d^{k_2}) \\ &\le C \cdot HOTT(d^{k_1}, d^{k_2})^{\frac{2}{15}}. \end{aligned}$$

6 Experiment

We follow the evaluation setup from Yurochkin et al. (2019) to assess LOTT's performance on k-NN classification. An LDA model with 70 topics is trained using a Gibbs sampler with 1500 iterations (Griffiths and Steyvers, 2004). To ensure consistency with HOTT, topics in LOTT are represented by the 20 most heavily-weighted words, renormalized to form proper probability distributions. The distance between topics in LOTT is computed using WMD, formulated as a standard linear programming problem and solved using the Gurobi optimizer and Python Optimal Transport (Gurobi Optimization, LLC, 2024; Flamary et al., 2021).

Table 1: Dataset statistics for evaluation

DATASET	|D|	V	AVG(W)	CLASSES
BBCSPORT	737	3657	116.5	5
TWITTER	3108	1205	9.7	3
OHSUMED	9152	8261	59.4	10
CLASSIC	7093	5813	38.5	4
REUTERS	7674	5495	35.7	8
AMAZON	8000	16753	44.3	4

6.1 Datasets

We consider 6 document classification benchmark datasets that are used to assess WMD and HOTT distances: BBC sports articles labeled with five sports (BBCSPORT), twitter posts labeled with positive, negative and neutral sentiments (TWITTER), medical records labeled with ten different cardiovascular disease groups (OHSUMED), academic paper segments labeled with four different publishers (CLASSIC), a standard news article dataset labeled with 8 distinct topics (REUTERS), and Amazon online reviews labeled with 4 product categories (AMAZON). All the documents are preprocessed such that stopwords and punctuation are removed. For each dataset, we split the data into 75% for training and 25% for testing. Table 1 provides key statistics for each dataset, including the number of documents |D|, the size of vocabulary V, the average unique words per document AVG(w), and the total number of classes.

We use the six datasets common to both Kusner et al. (2015) and Yurochkin et al. (2019), excluding 20NEWS (inconsistently constructed across the two studies) and GUTENBERG (not publicly available).

6.2 Document Distance Evaluation

We focus primarily on evaluating LOTT in comparison to several other optimal transport-based document distances. Our comparisons include Relaxed Word Mover's Distance (RWMD) (Kusner et al., 2015), a computationally efficient alternative to WMD that retains similar performance, as well as WMD-20, a truncated version of WMD that keeps only the top 20 unique words per document.

We also include HOTT as a baseline model for comparison given its shared reliance on LDA and GloVe embeddings (Pennington et al., 2014). We additionally evaluate a direct LDA embedding baseline, in which raw LDA topic proportion vectors serve as document representations and k-NN classification is performed using Euclidean distance, with no optimal transport applied. This allows us to isolate the contribution of the transport formulation over the topic representations; detailed results for which are reported in Table A.5 in the appendix. Again, we evaluate Hierarchical Optimal Full Topic Transport (HOFTT)(Yurochkin et al., 2019), a variant of HOTT without topic proportion truncation.

Since LOTT computes the linear optimal transport embedding based on a reference distribution σ, as a key hyperparameter in the model, we examine the effect of the reference using different choices for σ. To be specific, we experiment with 1, 2, 3, 4, 5, and 10 Gaussian mixtures to represent the reference distribution, analyzing how these choices influence the performance and computational efficiency of LOTT. The centers of the Gaussians are evenly spaced across the topic space. For a single mode, the center is placed at the middle of the topic vector, while for multiple modes, the centers are distributed equidistantly. All Gaussians share the same fixed Full-Width-Half-Maximum (FWHM) of 10 (see Figure A.1). To be precise, the reference is constructed over the one-dimensional discrete topic index space. For a reference with K components, the center of the i-th component is placed at index $\lfloor (i-1)\cdot(|T|-1)/(K-1) \rfloor$, spacing components evenly across the 70 topic indices. The FWHM controls the spread of each component in units of topic indices, and the resulting mixture is normalized to sum to one. A detailed sensitivity analysis of FWHM is provided in Table A.2.

6.3 Results

Computation. All embeddings and distance computations were implemented using Python 3.12 and executed on a machine with an Intel i9-13900HX 2.20 GHz processor and 32GB of RAM.

To evaluate computational efficiency, we measured the throughput, defined as the number of document pairs processed per second. This metric includes the time spent embedding all documents and performing pairwise comparisons in the k-NN algorithm. Higher throughput indicates better efficiency. Nevertheless, our results demonstrate that LOTT provides substantial improvements in computation speed across all datasets tested.

Table 2 presents throughput values for various methods, normalized relative to HOTT with the HOTT throughput set as 1.0 exactly. On small datasets like BBCSPORT, LOTT-5 achieves a speedup of 12.7× over HOTT. As the dataset size

Table 2: Standardized throughput of LOTT and other methods, normalized relative to the throughput of HOTT.

DATASET	WMD20	HOFTT	HOTT	LOTT-1	LOTT-5	LOTT-10	LOTT-15
BBCSPORT	0.374	0.307	1.000	11.467	12.683	12.524	12.593
TWITTER	0.397	0.290	1.000	98.579	101.807	101.073	101.646
OHSUMED	0.169	0.388	1.000	136.026	125.825	124.353	124.953
CLASSIC	0.275	0.212	1.000	93.333	87.735	86.156	84.983
REUTERS	0.149	0.177	1.000	92.032	91.417	89.976	89.441
AMAZON	0.074	0.287	1.000	182.299	168.049	168.388	165.111

and complexity grow, the advantages of LOTT become even more pronounced. For instance, LOTT-1 processes document pairs 136.0× faster than HOTT on the OHSUMED dataset and 182.3× faster on the AMAZON dataset, both of which feature larger vocabulary sizes and more documents.

The scalability of LOTT is further evidenced in datasets such as TWITTER and REUTERS, where all LOTT variants maintain consistently high throughput compared to HOTT, outperforming other methods like HOFTT and WMD20. For example, LOTT-1 achieves 98.6× the throughput of HOTT on the TWITTER dataset and 92.0× on the REUTERS dataset. A full breakdown of wall-clock time into preprocessing, topic cost computation, per-document embedding, and k-NN query stages is provided in Table A.3, showing that the offline preprocessing cost ranges from 74 seconds on TWITTER to 1634 seconds on AMAZON, while per-document inference remains below 21 seconds across all datasets.

Classification accuracy. We evaluate the classification performance of each OT-based method using a k-NN classifier with $k = 7$ for consistency (Figure 1). Despite the significant improvement in computation speed, our results show that LOTT achieves competitive classification accuracy compared to other OT-based document similarity metrics (Table A.1). For example, in the AMAZON dataset, LOTT-1 achieves a mean test error of 0.133, while LOTT-10 improves to 0.112, closely aligning with HOTT's 0.110. Meanwhile, LOTT-1 is on average 182× faster than HOTT, and LOTT-10 is 168× faster.

Figure A.2 presents the average test error across all datasets, normalized by nBOW. HOTT achieves a value of 0.63, while LOTT-10 reaches 0.65, indicating comparable performance between the two methods. Although LOTT's overall accuracy is slightly lower than HOTT, the difference is minor, and in some cases, LOTT performs better. For instance, on the TWITTER dataset, LOTT-4 achieves a mean test error of 0.316, outperforming HOTT (0.350) and the HOFTT variant (0.342). Similarly, for the OHSUMED dataset, LOTT and HOTT yield nearly identical results.

Parametrized method. An interesting observation is the impact of the reference distribution on both computation speed and classification accuracy. Figure A.3 highlights the trade-off between throughput and test error: as the number of Gaussian components in the reference distribution increases, the mean test error decreases, but the throughput also declines.

Among the LOTT variants, LOTT-10 achieves the lowest test error, demonstrating better classification performance. However, this comes at the cost of reduced throughput. Conversely, LOTT-1, which uses a single Gaussian component, is the fastest but exhibits higher test error. Beyond the number of components, the FWHM hyperparameter also actually affects accuracy on datasets with larger label spaces. On OHSUMED, varying FWHM from 10 to 1000 reduces test error from 0.54 to 0.45 while leaving throughput nearly unchanged, suggesting FWHM should be treated as a tunable parameter validated on a held-out split. A full sensitivity analysis across FWHM values is provided in Table A.2 and discussed in Appendix A.

We attribute LOTT-10's accuracy gains to its richer reference distribution, which better approximates the underlying corpus topic geometry and therefore aligns more closely with the full HOTT metric.

This parametrization enables LOTT to adapt to various tasks by allowing users to adjust the number of reference distributions. For instance, fewer references (e.g., LOTT-1) prioritize computational efficiency, while more references (e.g., LOTT-10) optimize classification accuracy. This flexibility makes LOTT a versatile tool, capable of balancing speed and accuracy.

Performance with limited training data. Figure A.4 illustrates how training data proportion

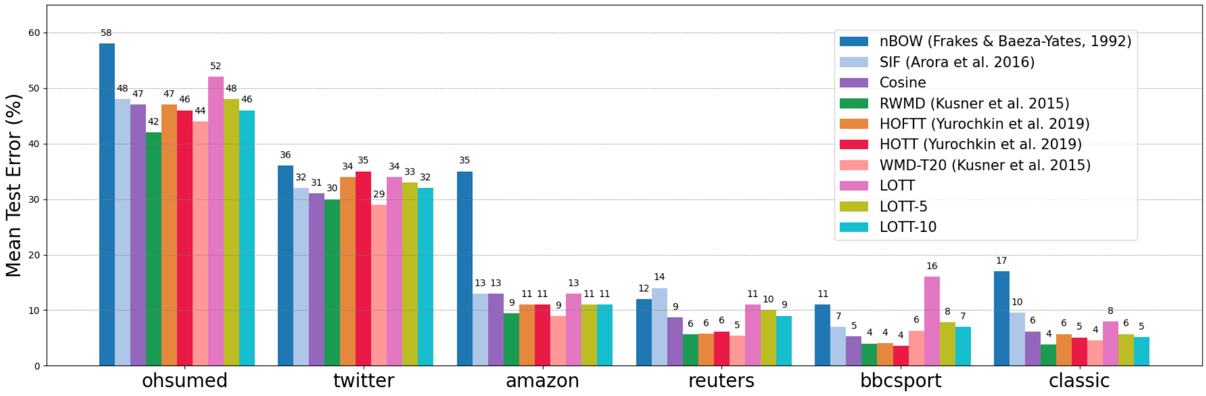

Figure 1: k-NN classification performance across datasets

affects mean test error in the CLASSIC dataset. Each line corresponds to a different training proportion (0.55 to 0.75) plotted against the number of reference distributions in LOTT.

LOTT remains robust across different training proportions when a suitable reference distribution is selected. Notably, LOTT-3 maintains consistently low test error regardless of training size, demonstrating its adaptability to limited data. This parametrization allows LOTT to be fine-tuned for optimal accuracy even with fewer training samples, making it a flexible alternative to fixed methods.

6.4 Complementarity with Lexical Methods

To assess whether LOTT captures information complementary to lexical similarity, we combined LOTT-10 distances with BM25 retrieval scores in a hybrid k-NN setting via linear interpolation. The hybrid consistently outperforms either method alone across all six datasets, with particularly large gains on BBCSPORT (test error falling from 0.119 to 0.027) and REUTERS (from 0.090 to 0.040). This indicates that LOTT's transport-based topic geometry captures semantic structure that BM25's lexical overlap does not, making the two methods complementary by construction. Full results for this study are reported in Table A.6 in the appendix.

6.5 Comparison to BERT

OT-based document similarity measures rely on classical statistical methods over bag-of-words representations, which differs fundamentally from transformer-based methods such as BERT that learn contextualized embeddings through self-attention mechanisms. To further investigate the performance of LOTT-10, we implemented BERT and its variants for comparison. We created document embeddings using SBERT, SBERT-LARGE, DistilBERT, RoBERTa, and BERT, and performed the same k-NN classification task using these embeddings on the datasets from our previous experiments. The specific checkpoints used here were `all-MiniLM-L6-v2` for SBERT, `all-mpnet-base-v2` for SBERT-LARGE, `distilbert-base-uncased` for DistilBERT, `roberta-base` for RoBERTa, and `bert-base-uncased` for BERT. It should be noted that for this set of BERT experiments, we utilized the original text versions of the datasets instead of the nBoW representations. In the previous experiments, we used nBoW representations to evaluate the topic transport methods; however, BERT models require raw text to generate embeddings. Using the original text datasets was preferred over reconstructing texts from nBoW representations, as reconstruction cannot recover the original sentence structures and contexts. For fair comparison, all methods were run on the same CPU, with computation time including both embedding generation and classification.

As shown in Table A.4, SBERT-LARGE generally achieved the lowest test error, outperforming LOTT-10 by an average of 0.065. However, this marginal accuracy improvement came at substantial computational cost—SBERT-LARGE required on average 196× more computation time than LOTT-10 (computation times are normalized with LOTT-10 as baseline). Among the transformer variants, SBERT offered the best speed-accuracy trade-off, achieving 0.051 lower average test error than LOTT-10 while requiring 33× more computation time. For DistilBERT, RoBERTa, and BERT, LOTT-10 demonstrated comparable classification accuracy, with LOTT-10 attaining higher accuracy on several datasets. Across all encoder methods, LOTT-10 consistently demonstrated superior computational efficiency while maintaining competitive performance with state-of-the-art transformer-based approaches.

6.6 t-SNE Visualization

Visualizing document embeddings as point clouds provides insight into classification performance. Using t-SNE (van der Maaten and Hinton, 2008), we visualized the CLASSIC dataset under three scenarios: baseline embeddings where each document is represented in a nBOW, embeddings obtained after applying topic modeling and LOT with one Gaussian reference, and embeddings generated using HOTT, as in fig. 2.

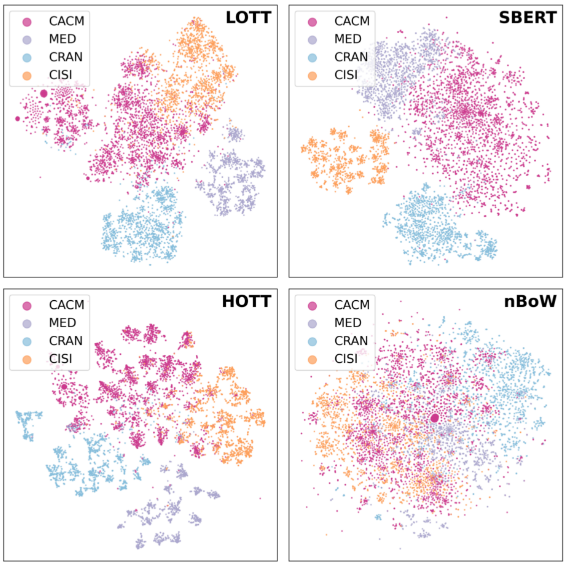

Figure 2: t-SNE on CLASSIC

The baseline plot shows minimal separation between classes, reflecting the limitations of nBOW in capturing semantic structure. In contrast, LOT, HOTT, and BERT embeddings show clearer class separation compared to the baseline. LOT embeddings exhibit distinct and homogeneous clusters, providing strong intra-class consistency. HOTT distances, while achieving tighter inter-class separation due to its hierarchical structure, show less homogeneity within clusters of the same class. This dispersion within classes suggests that HOTT may struggle to fully preserve intra-class relationships, potentially diluting its effectiveness for tasks requiring clear group cohesion. SBERT exhibits a more diffuse structure with substantial inter-category mixing, particularly between CACM and MED in the upper region of the projection.

7 Conclusion

In this paper, we introduced the Linear Optimal Topic Transport (LOTT) method, a novel approach to document classification that leverages linearized optimal transport embedding. LOTT achieves significant improvements in computation speed while maintaining competitive classification accuracy compared to traditional OT-based methods such as HOTT and WMD. This makes LOTT both scalable and efficient for real-world, large-scale applications. By embedding documents into a Euclidean space, LOTT transforms the complexity of pairwise optimal transport computations into efficient linear operations.

LOTT achieves up to 182× speedup over HOTT, is 33× faster than SBERT at only 0.051 higher average test error, and outperforms RoBERTa on five of six datasets. Combined with BM25, it further reduces error substantially across all datasets, confirming its complementarity with lexical methods.

LOTT's parametrized reference distribution offers a direct speed–accuracy tradeoff: LOTT-1 prioritizes throughput while LOTT-10 better approximates the Wasserstein geometry, making the method adaptable across data regimes including limited training data settings.

For future work, we would try to explore theoretical bounds on LOTT's performance in relation to WMD, as well as its potential applications in domains beyond document classification, such as image analysis or multimodal data comparison. LOTT's contribution is situated within the family of OT-based document similarity methods: it reduces pairwise transport computations to a linear number of offline embeddings without sacrificing classification accuracy relative to prior OT baselines. The linearization framework is agnostic to the choice of topic model, and replacing LDA with neural topic models such as BERTopic (Grootendorst, 2022) or contextual embedding-based topic representations (Zhang et al., 2022) is a natural direction for future work that could narrow the accuracy gap with transformer-based approaches while retaining LOTT's computational advantages.

Limitations

LOTT's performance depends on the quality of the underlying LDA topic model; when topic structure is weak or documents are very short, the transport-based embedding provides limited benefit over simpler baselines. The method also introduces several hyperparameters, including the number of LDA topics, the number of Gaussian components, and the FWHM parameter, each of which may require tuning. The scalability experiments here are conducted on corpora of a few thousand documents; while the theoretical complexity reduction holds at any scale and LOTT embeddings are compatible with approximate nearest neighbor indices for large-scale retrieval, full empirical validation at corpus sizes of hundreds of thousands of documents remains future work. Finally, LOTT operates on bag-of-words representations and does not capture word order or contextual information; its contribution is best understood as an improvement within the topic-transport family of methods rather than as a general alternative to contextual embeddings.

Beyond classification, LOTT shows promise for retrieval-augmented generation pipelines, where its efficiency in document retrieval directly impacts system responsiveness; initial experiments on TREC-COVID are discussed in Appendix C. Its foundation in optimal transport also provides interpretability that purely learned embeddings lack: document similarities can be traced back to topic distributions and the geometric relationships preserved by the embedding, which is valuable in high-stakes domains such as legal discovery or medical literature search. Deployment in such settings should be accompanied by appropriate safeguards in data handling and preprocessing.

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A Additional Tables

This appendix collects supplementary quantitative results that support the claims made in the main text. Table A.1 reports the full mean test error for all LOTT variants and OT baselines across datasets and references, extending the summary results discussed in Section 6.3. Table A.2 provides the FWHM sensitivity analysis for the OHSUMED and REUTERS datasets, showing that FWHM materially affects accuracy but leaves throughput nearly invariant. Table A.3 breaks down end-to-end wall-clock time into its constituent stages, separating one-time preprocessing costs from per-document inference costs. Table A.4 reports the full BERT model comparison with per-dataset test error and normalized inference time. Table A.5 reports the LDA embedding baseline, confirming that the OT formulation provides gains beyond topic modeling alone. Table A.6 reports the BM25+LOTT hybrid results, demonstrating the complementarity of transport-based and lexical similarity.

Sensitivity of FWHM

Table A.2 shows that the choice of FWHM has a meaningful effect on predictive accuracy for datasets with larger label spaces, while leaving computational throughput nearly unchanged. On OHSUMED (10 classes), test error decreases from 0.54 at FWHM=10 to 0.45 at FWHM=1000, a reduction of roughly 17%. On REUTERS (8 classes), the effect is smaller, with FWHM=50 attaining the lowest test error of 0.089. The throughput remains stable across all FWHM values on both datasets, confirming that the accuracy benefit comes at no computational cost.

The FWHM parameter controls the spread of each Gaussian component in units of discrete topic indices, determining how much probability mass from the reference distribution overlaps across neighboring topics. Narrow references (low FWHM) concentrate mass tightly around each component center, which can produce sparse transport plans that fail to capture gradual topic transitions in documents. Wider references provide richer overlap and therefore more expressive embeddings. A practical default heuristic is to set FWHM proportional to the inter-center spacing, FWHM $\approx |T|/K$, which provides sufficient overlap between components without excessive smoothing. For datasets with smaller label spaces and less diverse topic usage, the default FWHM of 10 used throughout the main experiments is generally adequate.

B Additional Visualizations

This appendix provides supporting visualizations for the experimental results in the main text. Figure A.1 shows the Gaussian mixture reference distributions used to construct the LOT embeddings, illustrating how probability mass is distributed across the 70 topic indices for one, two, five, and ten components. Figure A.2 presents the aggregated k-NN classification performance across all datasets, normalized by the nBOW baseline, providing a single-figure summary of the relative standing of all methods. Figure A.3 characterizes the speed-accuracy tradeoff as the number of reference distributions increases, showing that test error de-

Table A.1: Mean test error across datasets and references

DATASET	LOTT-1	LOTT-2	LOTT-3	LOTT-4	LOTT-5	LOTT-10	LOTT-15	HOFTT	HOTT
BBCSPORT	0.164	0.139	0.132	0.098	0.079	**0.070**	0.081	0.041	**0.036**
TWITTER	0.342	0.338	0.326	**0.316**	0.331	0.321	0.326	**0.342**	0.350
OHSUMED	0.523	0.525	0.489	0.491	0.478	**0.462**	0.464	0.469	**0.462**
CLASSIC	0.082	0.066	0.055	0.062	0.057	**0.053**	0.079	0.057	**0.051**
REUTERS	0.113	0.105	0.106	0.097	0.103	**0.093**	0.096	**0.058**	0.062
AMAZON	0.133	0.113	0.125	0.115	0.114	**0.112**	0.114	0.112	**0.110**

Table A.2: Sensitivity analysis of FWHM for Gaussian mixture reference distributions. We evaluate the impact of the FWHM hyperparameter, which scales the variance of a fixed 10-modal Gaussian mixture. We choose the OHSUMED (10 classes) and REUTERS (8 classes) datasets, which have larger label space. Computation times for each inference stage—including LOT embedding and kNN classification—are reported in seconds, while throughput is calculated as the ratio of test-train pairs processed to total inference time (pairs/s). The results show that while the choice of FWHM may impacts predictive accuracy—notably reducing the OHSUMED test error from 0.54 (FWHM=10) to 0.45 (FWHM=1000)—the computational overhead remains nearly invariant.

Dataset	**FWHM**	**Test Error**	**LOTT Embed (s)**	**kNN Fit (s)**	**kNN Pred (s)**	**Inference (s)**	**Throughput**
OHSUMED	10	0.5398	16.8739	1.1401	3.5337	21.5476	728,842.03
	50	0.4795	17.0449	1.1883	2.8443	21.0775	745,100.77
	100	0.4646	16.8072	0.7941	2.8823	20.4836	766,704.22
	500	0.4563	16.8260	0.7852	2.9194	20.5307	764,945.39
	1000	0.4471	16.7097	0.7672	2.9421	20.4189	769,130.60
REUTERS	10	0.0990	13.8689	0.7370	2.2017	16.8076	657,075.87
	50	0.0891	13.9693	0.7593	1.9977	16.7264	660,264.58
	100	0.0902	13.8191	0.7177	2.3485	16.8853	654,051.23
	500	0.0954	14.2413	0.7734	2.5377	17.5525	629,189.16
	1000	0.0933	13.8855	0.6827	1.9831	16.5513	667,250.02

Table A.3: Breakdown of end-to-end execution time. Setup costs (a and b) involve the one-time generation of latent topics and the computation of the topic-to-topic ground cost matrix. Inference costs (c and d) represent the per-document overhead for LOT embedding and subsequent kNN query. To ensure robustness, inference metrics are averaged across the full range of tested FWHM values.

Dataset	**(a) Prep/LDA**	**(b) Topic-Topic Cost**	**(c) LOTT-Embed**	**(d) kNN Query**	**Total (c+d)**	**Total E2E**
Twitter	73.76	0.78	6.30	0.93	7.23	**81.77**
BBC	174.00	0.98	1.33	0.16	1.49	**176.47**
R8	551.76	1.37	13.96	2.94	16.90	**570.04**
Classic	566.40	1.39	12.45	2.48	14.94	**582.73**
Ohsumed	1,073.11	2.21	16.85	3.95	20.81	**1,096.13**
Amazon	1,633.53	6.47	14.28	3.18	17.46	**1,657.46**

Table A.4: Test error and inference time of BERT models normalized by LOTT-10 performance. Inference time is expressed as a multiple of LOTT-10's time (baseline = 1.0). Both embedding generation and k-NN classification time are included for all methods.

DATASET	SBERT		SBERT-LARGE		DISTILBERT		ROBERTA		BERT	
	ERR	TIME	ERR	TIME	ERR	TIME	ERR	TIME	ERR	TIME
BBCSPORT	0.054	**40.909**	0.027	245.455	0.034	90.909	0.095	193.182	**0.014**	181.81
TWITTER	0.278	**2.352**	**0.268**	15.882	0.305	7.059	0.292	14.035	0.291	14.118
OHSUMED	0.315	**34.057**	**0.296**	181.682	0.503	70.421	0.551	137.313	0.513	149.389
CLASSIC	0.038	**20.312**	**0.021**	113.210	0.037	46.164	0.058	86.790	0.039	93.039
REUTERS	0.067	**24.301**	**0.059**	148.214	0.086	62.430	0.091	112.570	0.084	137.384
AMAZON	0.056	**42.631**	**0.049**	238.034	0.121	91.918	0.131	164.501	0.078	179.873

Table A.5: Mean test error and throughput of the LDA embedding baseline, where raw LDA topic proportions are used as document representations with k-NN classification via Euclidean distance. Throughput is normalized relative to LOTT-10. The results show that LOTT consistently outperforms the LDA baseline in both accuracy and throughput, confirming that the optimal transport formulation over topic proportions provides gains beyond the topic model alone.

DATASET	LDA TEST ERROR	REL. THROUGHPUT
BBCSPORT	0.065	1.656
TWITTER	0.336	0.422
OHSUMED	0.495	0.220
CLASSIC	0.058	0.172
REUTERS	0.072	0.260
AMAZON	0.131	0.215

Table A.6: Mean test error for LOTT-10 and the BM25+LOTT hybrid, where BM25 scores and LOTT distances are linearly interpolated. The hybrid consistently outperforms standalone LOTT-10, demonstrating that the two methods capture complementary sources of document similarity.

DATASET	LOTT-10	BM25+LOTT
BBCSPORT	0.070	**0.027**
TWITTER	0.321	**0.323**
REUTERS	0.093	**0.040**
AMAZON	0.112	**0.074**
CLASSIC	0.053	**0.036**
OHSUMED	0.462	**0.395**

creases monotonically while throughput declines gradually. Figure A.4 illustrates the robustness of LOTT to reduced training data on the CLASSIC dataset, showing that LOTT-3 achieves consistently low error across all tested training proportions from 0.55 to 0.75.

C LOTT for Retrieval-Augmented Generation Applications

Retrieval-Augmented Generation (RAG) is one of the most widely discussed topics in popular literature because of its capability of enhancing LLM generation outputs by retrieving relevant data and grounding the responses based on the fetched data. Therefore, we explored the capabilities of our novel LOTT method specifically in the retrieval part of RAG, experimenting whether the documents fetched from the corpus are actually relevant to the query document under consideration or not. We utilized Benchmarking Information Retrieval (BeIR) for this purpose, which is a standardized benchmark for evaluating the effectiveness of search and retrieval algorithms. We specifically focused on the TREC-COVID dataset, which is a collection of biomedical literature articles, in our studies here because of the diverse topic distributions provided across the documents in the dataset. This provides an opportunity for Topic Transport modeling to play along its strengths when compared against context-based embedding models like BERT. We conducted two different sets of experiments to compare the performance of LOTT against well-established BERT embeddings in RAG systems.

For the first set of experiments, we utilized the LOTT and BERT embeddings separately to create different FAISS vector datastores for saving the document corpus embeddings. Later, these stored FAISS indices were utilized to retrieve relevant documents corresponding to the query documents and various RAG-evaluation metrics like Precision@K, Recall@K, NDCG, MRR, etc. were computed to learn about the performance and utility of the two methods in RAG systems, where K is the number of retrieved documents. According to the results we obtained for this experiment, BERT embeddings outperformed LOTT embeddings in their application in RAG's retrieval step but our position in this regard is that the LOTT embedding method has a lot of hyperparameters that can be tuned to obtain better performance like the number of topics specified in the LDA model, which we have not explored extensively at the current moment and actually plan to explore in the future.

Similarly, for our second RAG-based experiment, we drew inspiration from (Shaaban et al., 2025), where the basic approach is to retrieve larger number of documents than are necessary using BERT embedding-based retrieval and later rerank these retrieved documents using LOTT embeddings to reach the set of most relevant documents for the queries. Again for the two-step approach, we utilized both LOTT embeddings based cosine similarity comparison and optimal transport based Word Mover Distance (WMD) comparison for finding the similar documents in the reranking phase. When we compared direct BERT retrieval results with this two-step method of BERT retrieval+LOTT reranking, we found that direct BERT retrieval performed better than our two-step approach. We anticipate superior results from this two-step retrieval+reranking process possibly through swapping the BERT and LOTT functions in the pipeline,

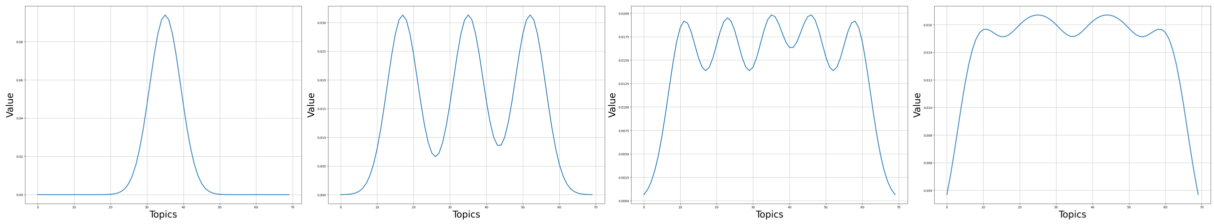

Figure A.1: Visualization of Gaussian mixture reference distributions used for LOTT embedding creation. Each subplot shows the probability mass distribution across topics for the corresponding number of Gaussian references.

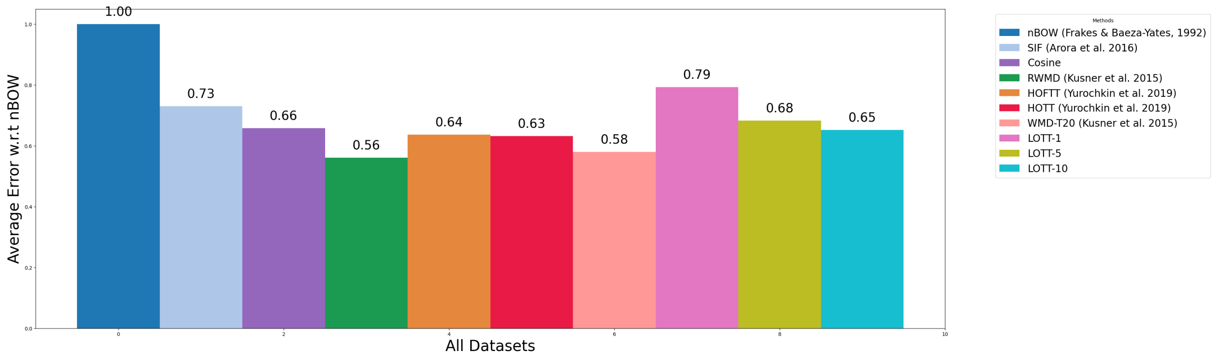

Figure A.2: Aggregated k-NN classification performance relative to nBOW baseline. All methods' test errors are normalized by dividing by nBOW's test error, where values below 1.0 indicate better performance than nBOW.

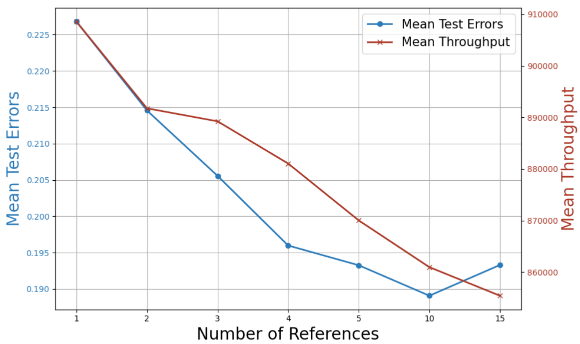

Figure A.3: Performance analysis of LOTT across different numbers of reference distributions. Mean test error and mean throughput averaged across all datasets as a function of the number of references. As the number of references increases, the LOT embedding provides a better approximation of optimal transport distances, with LOTT-10 using 10 reference distributions to balance expressiveness and computational efficiency.

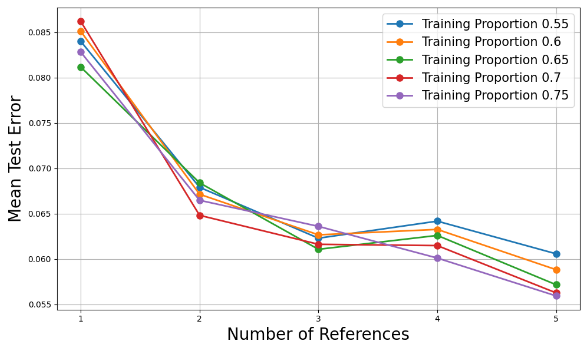

Figure A.4: Effect of training data proportion on LOTT performance. Mean test error on the CLASSIC dataset for training proportions from 0.55 to 0.75, plotted against the number of reference distributions. LOTT-3 demonstrates robust performance across all training sizes. Also, as training proportion decreases, test error generally decreases with increasing references, showing the method's effectiveness with limited training data.

i.e., retrieving a crude set of documents first using LOTT and then reranking using BERT, and/or hyperparameter tuning like changing the number of intermediate documents retrieved in the initial step. These are the future exploration directions we are mostly focused towards at the moment to demonstrate the capability of our novel LOTT method for information retrieval tasks.

D Empirical Validation of the Affine Pushforward Assumption

The LOT approximation bounds invoked in section 5 rely on the assumption that document topic distributions are approximately generated by shifting and scaling a shared reference distribution. To assess whether this holds empirically, we designed a comparative experiment between a restricted global generative model and an independent per-

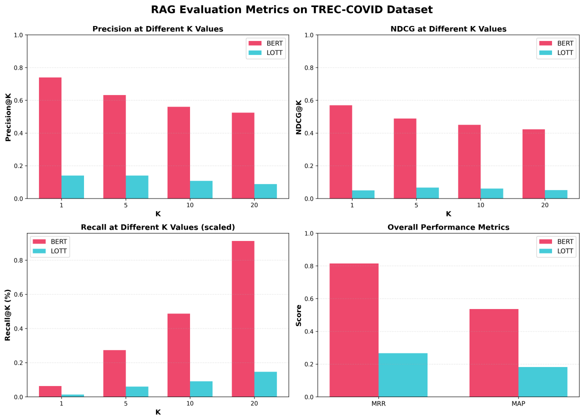

Figure A.5: RAG Metrics on TREC-COVID

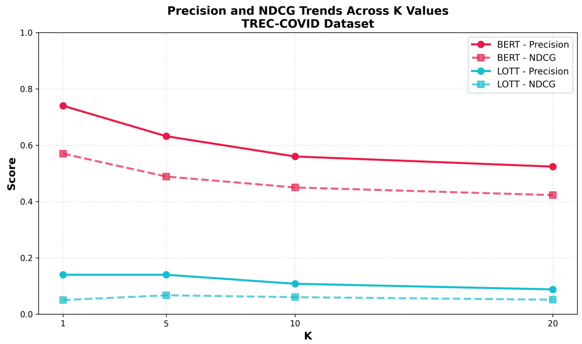

Figure A.6: Precision/NDCG Trends on TREC-COVID

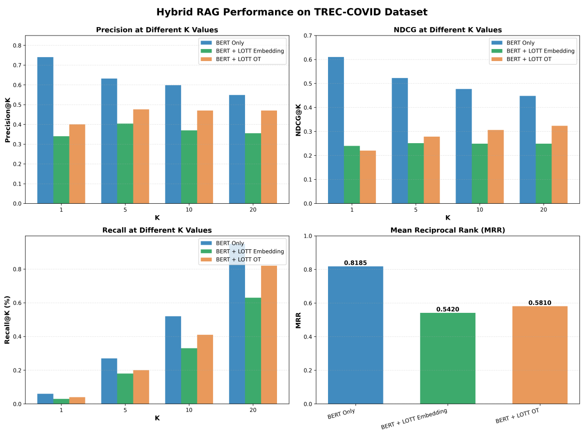

Figure A.7: Hybrid RAG Metrics on TREC-COVID

document baseline on the CLASSIC dataset.

For the restricted model, we pool Monte Carlo samples drawn from every document's empirical topic distribution into a single corpus-level bag and fit a 10-component GMM to this pooled sample. For each document, we then perform a grid search over shifting and scaling parameters and select the transformation that maximizes the log-likelihood of that document's empirical distribution under the transformed global GMM. The total log-likelihood

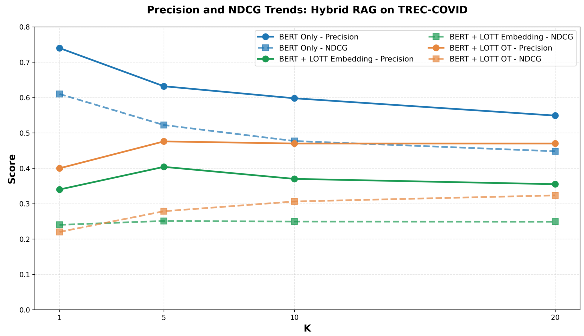

Figure A.8: Hybrid RAG Precision/NDCG Trends on TREC-COVID

is summed across all documents in the corpus.

For the independent baseline, we fit a separate GMM to each individual document using only that document's samples. Because these models are overfit to individual documents, this yields an upper bound on the achievable total log-likelihood under any per-document scheme.

On the CLASSIC dataset, the restricted model achieved a total log-likelihood of -2993 compared to -3683 for the independent baseline, a difference of $+690$ in favor of the restricted global model. The fact that a constrained model with a shared generative structure outperforms per-document overfit models provides empirical support for the hypothesis that topic distributions across a corpus share an affine structure. This finding offers a practical justification for the reference GMM construction used in LOTT and helps explain why the linearization preserves classification accuracy relative to the full pairwise HOTT computation.

E Deployment Regime and Scope of Comparison

The throughput comparisons in this work are conducted in a CPU-only setting, using an Intel i9-13900HX processor with 32GB of RAM for all methods including the BERT-family baselines. This choice reflects a concrete and practical deployment scenario: edge computing, on-premise enterprise systems without GPU infrastructure, and low-cost cloud inference instances represent a substantial fraction of real-world NLP deployments. LOTT's efficiency advantage is most pronounced in this regime.

Transformer-based models benefit substantially from GPU parallelization, and a GPU-inclusive throughput comparison would reduce LOTT's efficiency advantage relative to those methods. We intend to include GPU throughput figures in a future version of this work, reporting pairs per second for all BERT-family baselines on a standard single GPU alongside the CPU results, so that readers can assess the tradeoffs across both deployment settings.

We also note that LOTT's pipeline is not inherently tied to LDA or GloVe embeddings. The LOT embedding framework applies to any distributional document representation, and replacing LDA with neural topic models such as BERTopic (Grootendorst, 2022) or contextual clustering-based topic representations (Zhang et al., 2022) is a natural extension. LDA was chosen here for direct reproducibility and comparability with WMD (Kusner et al., 2015) and HOTT (Yurochkin et al., 2019), the two prior works this paper most directly extends.